Worksheet 8.6

Newton's Second Law

Student: ..... Class: .....

1. State Newton's Second Law of Motion.

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2. Use the formula $F = ma$ to complete the following table

Force (F) (newtons)	Mass (m)	Acceleration (a)
100	40 kg	
2	200 g	
70		3.5 m/s^2
100		10 cm/s^2

3. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.



(a) Calculate the net force acting on the body.

(b) Calculate the acceleration of the body.

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4. The photo shows a drag car coming to a stop at the end of a race.



Photo: Claudia Schnepf / 123RF

(a) To come to a complete stop, the drag car negatively accelerates. Identify another term for 'negative acceleration'. .....

(b) Identify the force that brings the drag car to a stop. .....

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